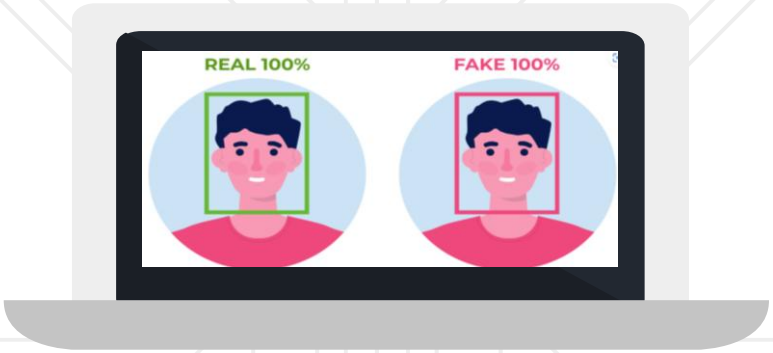
DeepFake
Image Detection using
Convolutional neural network

OBJECTIVE

To develop a classification model that best distinguishes Deepfake Images from Real Images.



Contents

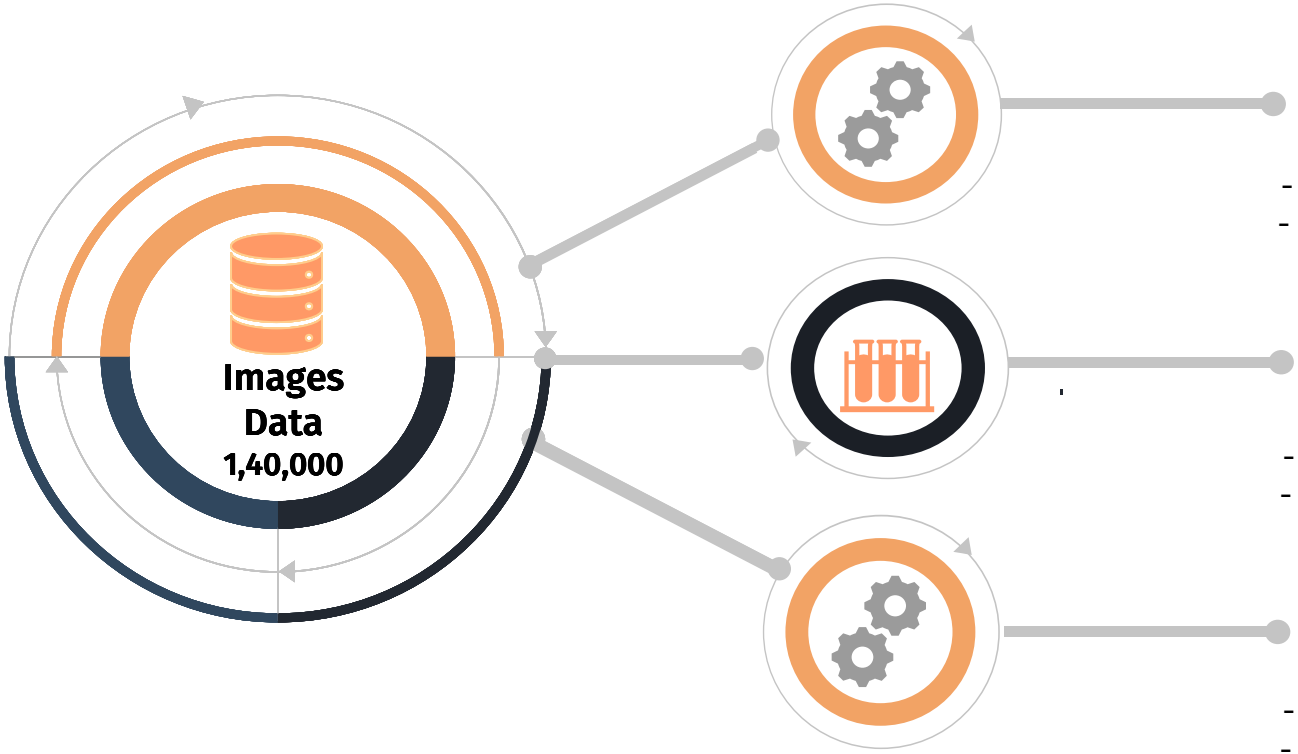
Data Description & Image Pre-Processing

Model Selection & Fitting

Performance Evaluation

Limitations & Future Scope

Data Description



Train (70%)

1,00,000 Images

- 50,000 Real Images
- 50,000 Fake Images

Validation (15%)

20,000 Images

- 10,000 Real Images
- 10,000 Fake Images

Test (15%)

20,000 Images

- 10,000 Real Images
- 10,000 Fake Images

1. Image Pre-Processing

Resize Images

Images need to be resized to a fixed size to model.

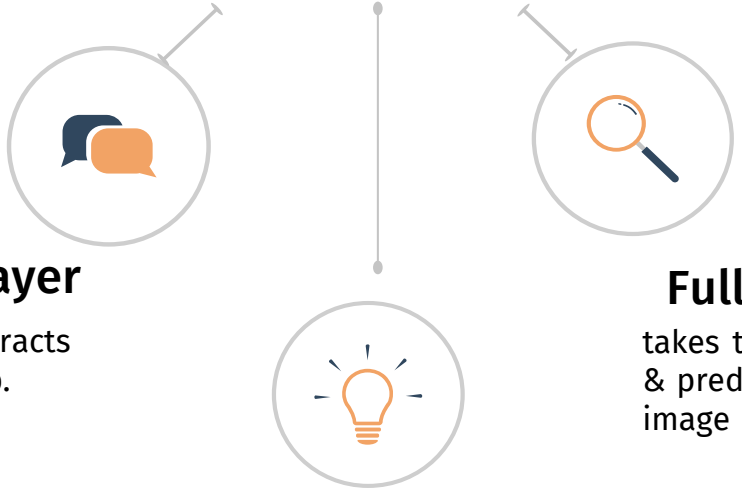
Normalize Images

Normalize pixel values in the range of (0,1)

Flatten Images

Convert 2D image to 1D array

Convolution Neural Network (CNN)



Convolution Layer

slides a filter over the image & extracts features resulting in a feature map.

Pooling Layer

performs down-sampling by reducing the size and sends only the important data to next layers in CNN.

Fully Connected Layer

takes the output of convolution/pooling & predicts the best label to describe the image

Convolution Layers of CNN

0	0	0	0	0	0	0
0	60	113	56	139	85	0
0	73	121	54	84	128	0
0	131	99	70	129	127	0
0	80	57	115	69	134	0
0	104	126	123	95	130	0
0	0	0	0	0	0	0

Input Image

0	-1	0
-1	5	-1
0	-1	0

Kernel

114				

Feature Vector

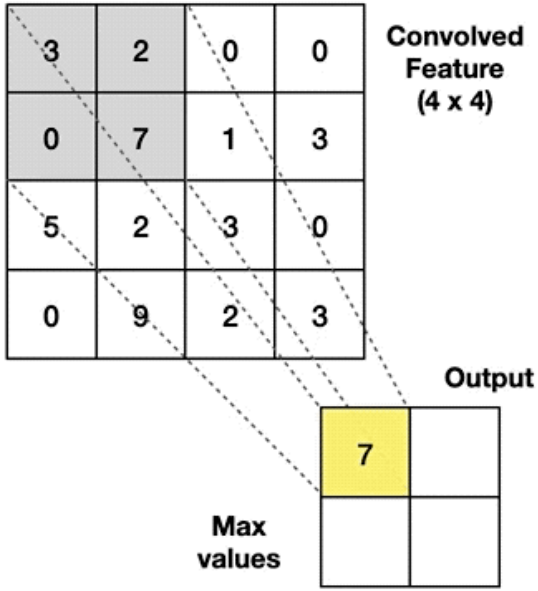
Padding in CNN
Padding is a process of adding layers of zeros to our input images so as to avoid the problem of image shrinkage.

Strides in CNN
The stride describes the step size when sliding the convolutional filter over the input image.

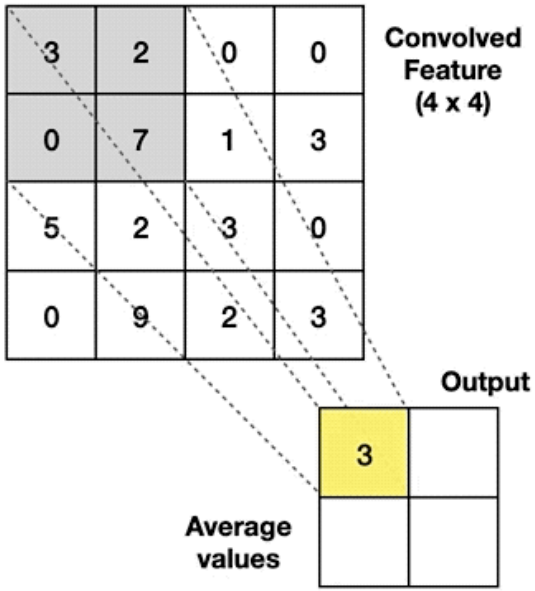
Pooling Layers of CNN

Pooling layers reduce the dimensions of the feature maps. It mainly helps in extracting sharp & smooth features.

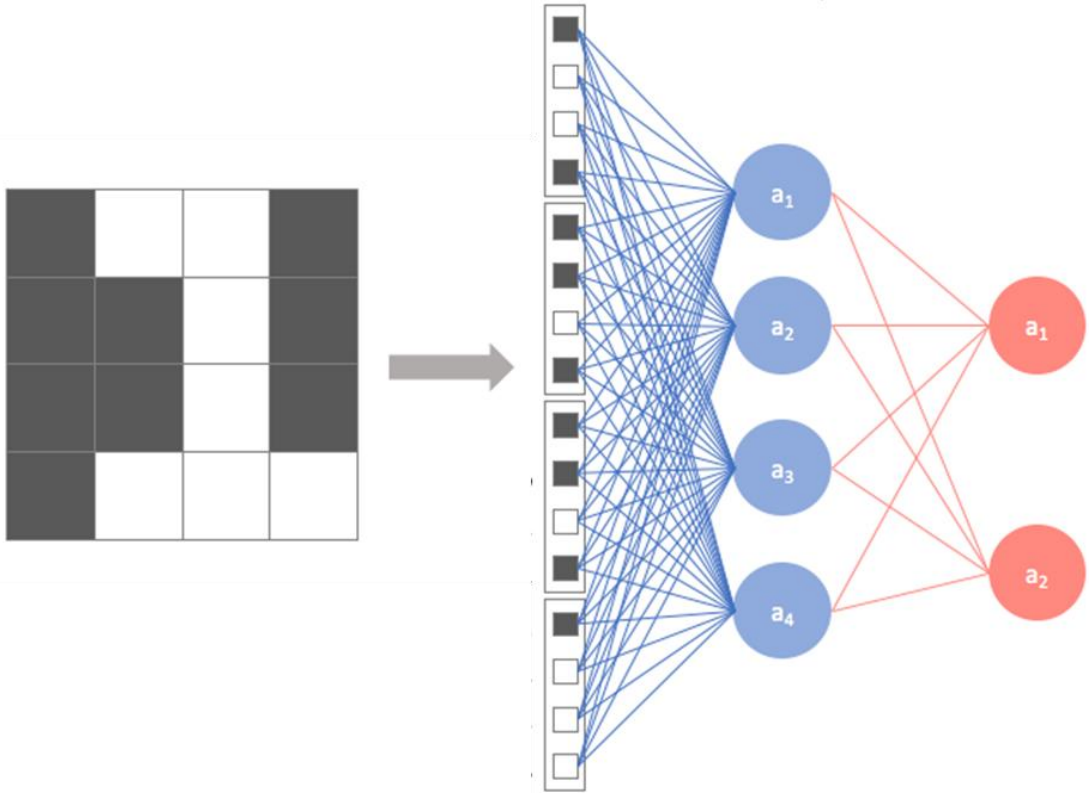
Max Pooling



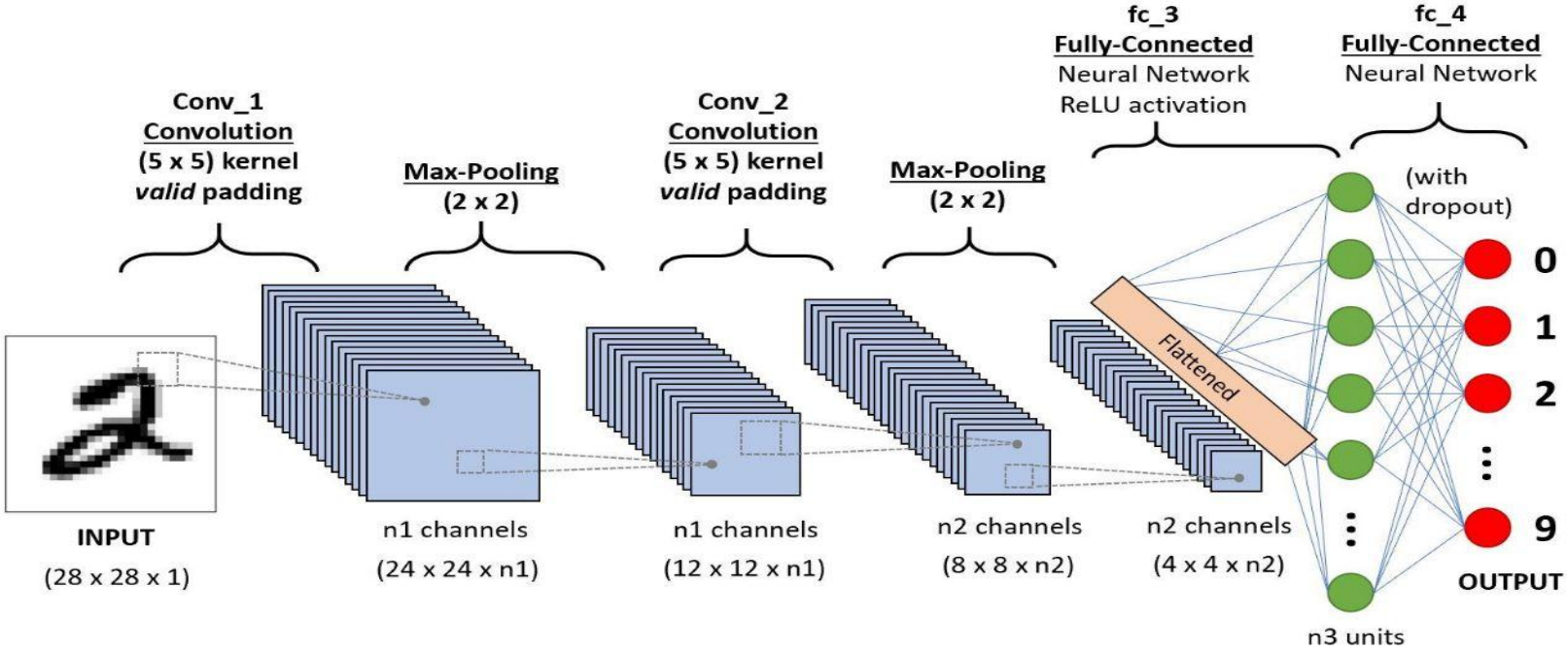
Average Pooling



Fully Connected Layers of CNN

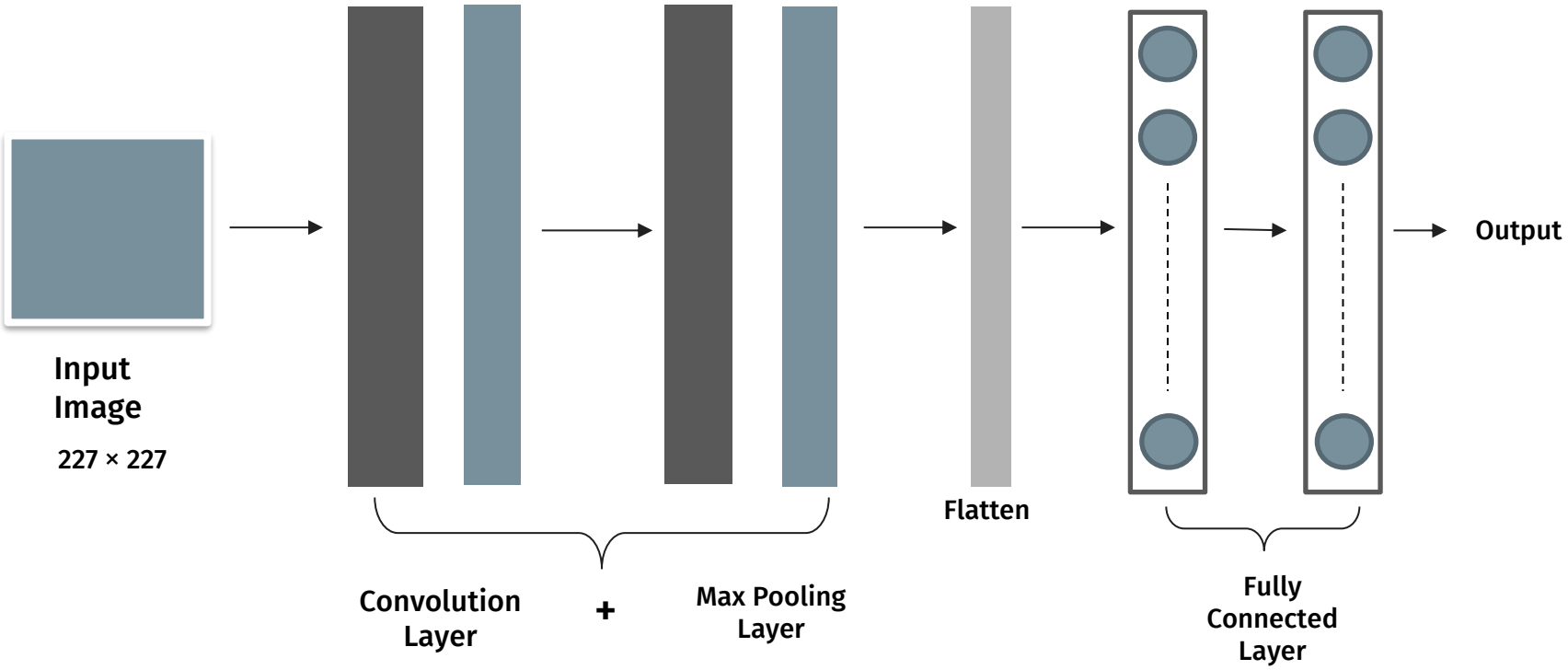


CNN Architecture



Custom CNN Models

Custom-I Architecture



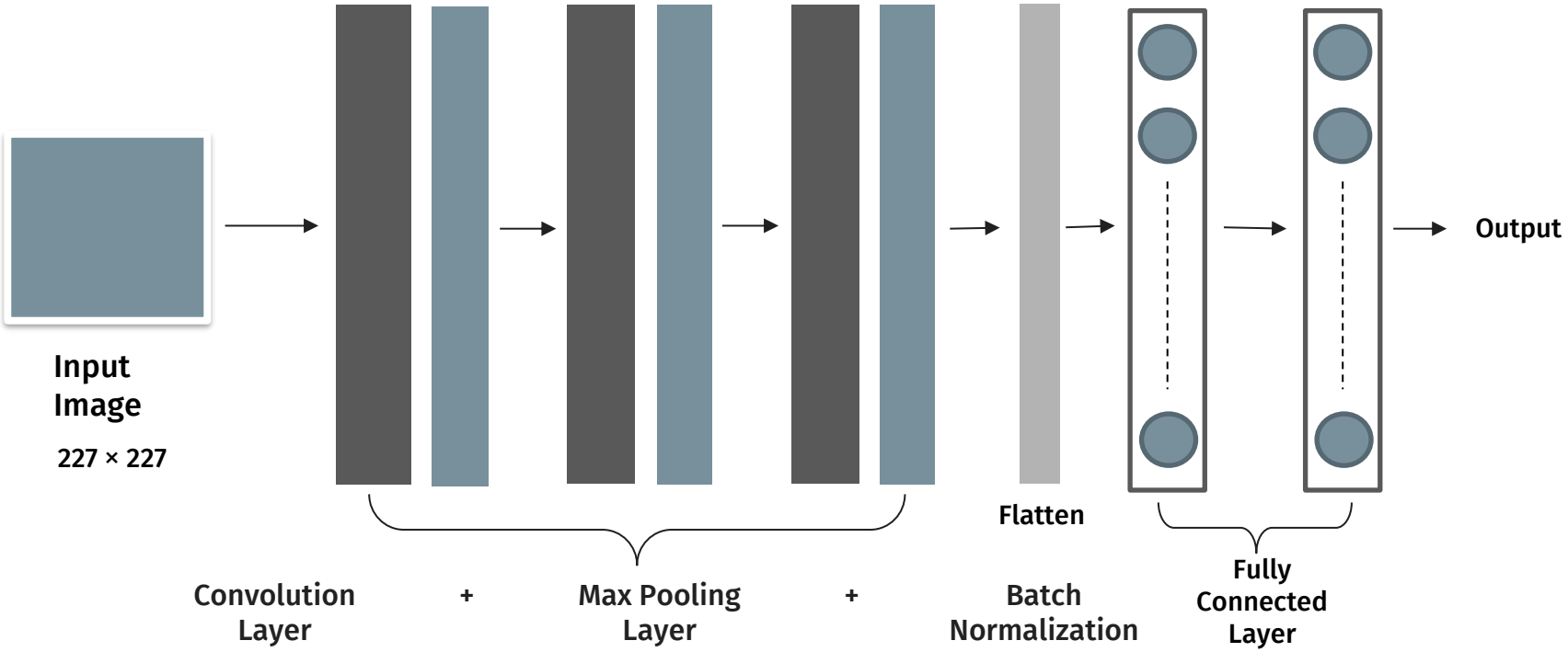
Custom-I Model Performance

	Accuracy	Recall	Precision	AUC-Score
Train	0.9914	0.9927	0.9902	0.9994

	Accuracy	Recall	Precision	AUC-Score
Validation	0.9409	0.9344	0.9468	0.9772

	Accuracy	Recall	Precision	AUC-Score
Test	0.9427	0.9355	0.9491	0.9788

Custom-II Architecture



Custom-II Model Performance

	Accuracy	Recall	Precision	AUC-Score
Train	0.9736	0.9777	0.9699	0.9970

	Accuracy	Recall	Precision	AUC-Score
Validation	0.9589	0.9574	0.9603	0.9911

	Accuracy	Recall	Precision	AUC-Score
Test	0.9598	0.9584	0.9611	0.9909

Challenges

Model training is computationally extensive

Lack of proper system infrastructure prohibited use of certain ML and DL models.

Future Scope

A DeepFake Video Detection to detect the DeepFake Videos

Using Input deepfake images having multiple faces.

Other CNN Pre-trained models can be tried.

References

1. Introduction To Convolutional Neural Networks | by Prince Krampah | Medium
2. https://www.mygreatlearning.com/blog/all-you-need-to-know-about-deepfake-ai/
3. https://towardsdatascience.com/illustrated-10-cnn-architectures-95d78ace614d#676b
4. CampusX- 100 days of deep learning

THANK YOU